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MILITARY SPECIFICATION

ADAPTER, MOUNT ASSEMBLY: 10541111

1. SCOPE

1.1 This specification covers one type of adapter mount which supports the Range Finder, Laser AN/GVS-3 and provides a scale for azimuth measurement (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance, and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Enviromental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F7691596	Fixture, Azimuth Test
F10541111	Adapter, Mount Assembly
F10544049	Gage, Interchangeability, Maximum
F10544050	Gage, Interchangeability, Minimum
F10544051	Gage, Interchangeability, Maximum
F10544052	Gage, Interchangeability, Minimum
B11730226	Connector

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PACKAGING DATA SHEETS

10541111 Packaging of Adapter, Mount Assy

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The adapter mount shall be manufactured in accordance with Drawing F10541111 and drawings pertaining thereto.

3.3 General specification.- The adapter mount shall meet the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Electrical and electronic assemblies
(d) Soldering electrical connections
(e) Part identification and marking
(f) Workmanship

3.4 Environmental.

3.4.1 Vibration.- The adapter mount shall withstand vibration for 15 minutes each in 3 mutually perpendicaular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequency. Subsequent to vibration, the adapter mount shall show no evidence of physical damage, and shall meet the requirements of 3.5 through 3.6.2.

3.4.2 Shock.- The adapter mount shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular directions. Each shock impulse shall be a half-sine wave with a time duration of 11 $\pm$ 1 milli-seconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the adapter mount shall show no evidence of physical damage and shall meet the requirements of 3.5 through 3.6.2.

3.4.3 Storage temperatures.- The adapter mount shall withstand thermal stabilization for a period of 5 hours at temperatures of -80° Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the adapter mount shall show no evidence of physical damage, and shall meet the requirements of 3.5 through 3.6.2.

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3.4.4 Rain.- The adapter mount shall be capable of withstanding 5 hours of continuous exposure to heavy rainfall in the amount of 2.0 inches/hour with a wind speed of 35 knots at a water temperature of plus 85°F. Subsequent to such rainfall, the electrical circuitry shall not have been damaged or adversely affected and the adapter mount shall meet the requirements of para. 3.5 through 3.6.2.

3.4.5 Humidity.- The adapter mount shall be capable of withstanding 4 hours of continuous exposure to a relative humidity of 90 to 95 percent at a temperature of +155°F ± 2°F. During and subsequent to the above humidity, the adapter mount shall not have been damaged or adversely affected, and shall meet the requirements of paragraph 3.5 through 3.6.2.

3.4.6 Sand and Dust.- The adapter mount shall be capable of withstanding 3 hours of continuous exposure to blowing sand of particle size 0.01 to 1.00 mm and blowing dust of particle size 0.001 to 0.01mm with a density between 0.1 to 0.5 gram per cubic foot of sand and a density between 0.1 to 0.5 gram per cubic foot of dust at a temperature of plus 75°F, a humidity between 15 to 30%, and a minimum wind velocity of 1500 ± 100 feet per minute. Subsequent to sand and dust exposure, the adapter mount shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.5 through 3.6.2.

3.4.7 Salt Spray.- The adapter mount shall be capable of withstanding 48 hours of continuous exposure to a fine spray of 5% salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a PH between 6.5 and 7.2, and a temperature between plus 93°F and plus 97°F. Subsequent to salt spray, the adapter mount shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.5 through 3.6.2.

3.4.8 Fungus.- The adapter mount shall be capable of withstanding 28 days exposure to a composite spore suspension consisting of the following cultures: (a) Chaetomium globosum, (b) Aspergillus versicalor, (c) Aspergillus flavus, (d) Pencillium luteum, (e) Rhizopus nigricans. The composite spore suspension shall be incubated at an ambient temperature of between plus 77°F and plus 84°F with an RH of 97 ± 2%. Subsequent to the above fungus exposure, the adapter mount shall not have been damaged or adversely affected and shall meet the requirements of paragraph 3.5 through 3.6.2.

3.4.9 Operational temperatures.- The adapter mount shall meet the requirements of 3.5.1 through 3.6.2 when operated at a thermally stabilized condition at temperatures of -65°F, +155°F and at a room temperature of between +70° and +80°F. The adapter mount shall also meet the requirements of 3.5.1 through 3.6.2 for all temperatures between -65° and +155°F.

3.5 Operability.- The adapter mount shall meet all functional requirements of this specification when the following representative components of the Range Finder AN/GVS-3 are employed.

3.5.1 Adapter mount-to-shelf mount.- The adapter mount shall be capable of being seated on a shelf mount, and locked in position, when the locking lever of the adapter mount is moved to the locked position. When the locking lever is in the unlocked position (maximum counterclockwise position), the adapter mount shall be capable of being removed, without restraint, from the shelf mount.

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3.5.2 Adapter mount-to-yoke assembly.- The adapter mount shall be capable of accepting and seating a yoke assembly and securely locking it into position. Electrical continuity shall exist between the electrical contacts of the adapter mount is rotated with a torque specified in 3.5.3 through its maximum clockwise excursion. When the locking knob is rotated through its maximum counterclockwise excursion, the yoke assembly shall be capable of being removed, without restraint, from the adapter mount.

3.5.3 Locking torque (yoke assembly-to-adapter mount).- The torque required to rotate the locking knob of the adapter mount, through maximum clockwise excursion, to secure a yoke assembly to the adapter mount shall be 10-12 inch-pounds.

3.5.4 Unlocking torque (yoke assembly-from-adapter mount).- The maximum torque required to rotate the locking knob of the adapter mount through maximum excursion, to unlock and release a yoke assembly from the adapter mount shall not exceed 12 inch-pounds.

3.5.5 Azimuth excursion.- The azimuth excursion through which the adapter mount shall move a yoke assembly, secured to the adapter mount, shall be at least 5675 mils.

3.5.6 Azimuth error.- The error in azimuth scale readings taken in 10 mil steps throughout the total excursion (6400 mils) of the azimuth scale shall not exceed 2.0 mils. The total spread between the largest positive error and the largest negative error shall not exceed 4.0 mils.

3.5.7 Circular level.- The level bubble of the adapter mount shall remain within the inner circle of the level, when a yoke assembly is moved in azimuth through an excursion of 5675 mils.

3.5.8 Azimuth knob rotation.- The maximum torque required to rotate the azimuth adjustment knob for azimuth rotation of a yoke assembly shall not exceed 30 inch-ounces when thermally stabilized at a temperature between 70° to 80°F, and shall not exceed 60 inch-ounces when the adapter mount is thermally stabilized at the extreme temperatures of 3.4.9.

3.5.9 Plate rotation (normal operating temperature).- The minimum torque required for rotation of Plate 10541125 shall be not less than 15 inch-pounds, when the plate latch is in its normal position and the adapter mount is subjected to a temperature between 70° to 80°F.

3.5.10 Plate rotation (extreme operating temperatures).- The maximum torque required for rotation of Plate 10541125 shall not exceed 12 inch-pounds, when the plate latch is fully depressed and the mount is subjected to the extreme temperatures of 3.4.9.

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3.5.11 Leveling adjustment screws.- The maximum torque required to rotate each of the three leveling adjustment screws of the adapter mount shall not exceed 4 inch-pounds when the adapter mount is thermally stabilized at a temperature between 70° to 80°F, and shall not exceed 7 inch-pounds when subjected to the extreme temperatures of 3.4.9.

3.5.12 Cable assembly.- The cable assembly of the adapter mount (for connecting the adapter mount to the battery of the range finder) shall withstand bending through an angle of at least 270 degrees around a 1/2 inch diameter mandrel, when thermally stabilized at the temperatures of 3.4.9. There shall be no evidence of physical damage or electrical failure.

3.5.13 Spindle rotation (normal operating temperature).- The torque required for rotation of spindle 10541117 shall be 12-16 inch-pounds, when the spindle is in its central position, and the adapter mount is thermally stabilized at a temperature between 70°F to 80°F.

3.5.14 Connector interface requirement.- The connector shall interface with Connector 11730226 (22J1) of battery pack assembly.

3.6 Electrical.- The electrical wiring of the adapter mount shall conform to the schematic drawing.

3.6.1 Continuity.- Each conductor of the cable shall be electrically continuous (nominally 0 resistance) between the appropriate terminations as specified on the electrical wiring diagram F10541111.

3.6.2 Insulation resistance.- The insulation resistance from each pin to every other pin and the the connector body shall not be less than 1000 megohms, when subjected to a minimum potential of 500 volts, direct-current.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 mounts plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production."

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
101.	Adapter mount to shelf mount	3.5.1	4.6.2
102.	Adapter mount to yoke assembly	3.5.2	4.6.3
103.	Locking torque (yoke assembly to adapter mount)	3.5.3	4.6.4.1
104.	Unlocking torque (yoke assembly from adapter mount)	3.5.4	4.6.4.2
105.	Azimuth knob rotation	3.5.8	4.6.4.3
106.	Plate rotation	3.5.9	4.6.4.4
107.	Leveling adjustment screws	3.5.11	4.6.4.5
108.	Spindle rotation	3.5.13	4.6.4.7
109.	Azimuth excursion	3.5.5	4.6.5.1
110.	Azimuth error	3.5.6	4.6.5.2
111.	Circular level	3.5.7	4.6.5.3
112.	Continuity	3.6.1	4.6.6.1
113.	Insulation resistance	3.6.2	4.6.6.2
114.	Interface requirement	3.5.14	4.6.7

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between ≠70° and ≠80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- Three mount assemblies shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable drawings & Specs - Visual
302.	Fabrication	3.2	Applicable drawings - Visual
303.	General specification	3.3	MIL-F-13926 - Visual

NOTE: The tests of Table II shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three mount assemblies shall be selected at random as a special sampling from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then be subjected to the tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration (+70° and +80°F)	3.4.1	4.6.1.3
305.	Shock (+70° and +80°F)	3.4.2	4.6.1.2
306.	Storage temperature (-80° and +160°F)	3.4.3	4.6.1.1
307.	Rain	3.4.4	4.6.1.4
308.	Humidity	3.4.5	4.6.1.5
309.	Sand and Dust	3.4.6	4.6.1.6
310.	Salt Spray	3.4.7	4.6.1.9
311.	Operational temperatures (-65°F and +155°F)	3.4.9 through 3.6.2	4.6.1.7, 4.6.2 through 4.6.7

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Adapter mount accuracy.- The special testing equipment utilized in testing the accuracy of the adapter mount shall consist of the following:

a. Yoke assembly.- The device representing the yoke assembly shall consist of a viewing telescope, microscope and a support (See Figure 1). Provision shall be made in the support for positioning the line of sight of the telescope parallel to base, and to intersect the axis of the Adapter Mount Spindle (Dwg. F10541111); alignment of telescope reticle image to the target collimator reticle image; mounting the support to the spindle of adapter mount; and positioning the microscope to align the microscope reticle image to the azimuth scale graduation of the adapter mount.

b. Adapter.- The device representing the adapter shall provide for the mounting of the adapter mount to an azimuth test fixture and alignment of the adapter mount spindle axis to the axis of the rotary table of the azimuth test fixture. The azimuth test fixture Drawing F7691596 may be utilized as a guide for set-up of the testing equipment.

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c. Target collimator.- A target projector collimator with a clear aperture of at least 1.5 inches and an appropriate reticle pattern for measuring the mechanical accuracies of the adapter mount shall be part of the test equipment. The optical axis of the target projector collimator shall be positioned parallel to the azimuth test fixture table and intersect the axis of rotation. A set-up device (of the front surface reflecting mirror type) shall be utilized for the alignment (auto-collimation) of the target projector collimator. The vertical reticle line of the target collimator shall be adjusted plumb.

d. Accuracy.- The special testing equipment utilized for the azimuth excursion test shall be within 1 mil and for the azimuth error test shall be within .5 mil.

4.5.2.3.2 Adapter mount to shelf mount, positioning.- The special testing equipment utilized in this test shall conform to Drawings F10544051 and F10544052.

4.5.2.3.3 Adapter mount to yoke assembly positioning and locking torque.- The special testing equipment utilized in these tests shall conform to Drawings F10544049 and F10544050.

4.6 Test methods and procedures.

4.6.1 Environmental.- The adapter mount shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the adapter mount shall be subjected to the tests specified in 4.6.2 through 4.6.7.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.4.3.

4.6.1.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.4.2.

4.6.1.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The adapter mount shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4.1. The vibration test shall then be performed in accordance with Method 510 Procedure I, Part 1 of MIL-STD-810.

4.6.1.4 Rain.- The rain test shall be performed as outlined in MIL-F-13926, Procedure I with the following exceptions: Five (5) hours of continuous rainfall

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at a rate of 2 inches/hour and wind velocity of 35 knots. Temperature of the water shall be maintained at approximately 85°F.

4.6.1.5 Humidity.- The humidity test shall be performed as outlined in MIL-F-13926, Procedure II and under the conditions specified in 3.4.5 for four (4) hours.

4.6.1.6 Sand and dust.- The sand and dust test shall be performed as outlined in MIL-F-13926, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.4.6.

4.6.1.7 Operational temperatures.- The test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.4.9 utilizing the test equipment indicated in 4.6.1.1. The adapter mount shall be exposed to each specified temperature for a minimum period of 5 hours. At the end of each 5 hour period, and while still at the specified temperature, the adapter mount shall be subjected to the tests in 4.6.2 through 4.6.7, to determine conformance to the applicable requirements of 3.5 through 3.6.2.

4.6.1.8 Fungus.- One (1) adapter mount shall be subjected to the fungus test outlined in MIL-F-13926. At the end of the 28 day incubation period, the adapter mount shall be tested as specified in 4.6.2 through 4.6.7 for compliance with 3.5 through 3.6.2. Subsequent acceptance of the adapter mount shall be by certification.

4.6.1.9 Salt spray.- The salt spray test shall be performed as outlined in MIL-STD-810, Method 509, and according to the conditions specified in 3.4.7.

4.6.2 Adapter mount to shelf mount.- Perform this test utilizing the special testing equipment specified in 4.5.2.3.2. The adapter mount shall be properly secured to each shelf mount gage when the locking lever of the adapter mount is moved to the locked position as specified in 3.5.1. Subsequent to the above test move the locking lever to the unlocked position (maximum counter clockwise position). Repeated installation of the adapter mount to each shelf mount gage shall be accomplished, a minimum of three times. Observe for any evidence of binding or interference at each installation. The adapter mount shall meet the requirements specified in 3.5.1.

4.6.3 Adapter mount to yoke assembly.- Perform this test with the special testing equipment specified in 4.5.2.3.3 positioned on the adapter mount. Utilizing standard measuring equipment, electrical continuity shall be determined when the locking knob of the adapter mount is rotated throughout its maximum clockwise excursion. Rotate the locking knob through its maximum counterclockwise excursion. Repeated installation of each yoke assembly gage to the adapter mount shall be accomplished, a minimum of three times. Observe for any evidence of binding or interference at each installation. The adapter mount shall meet the requirements specified in 3.5.2. Continuity shall be checked with standard measuring equipment during rotation of the adapter mount.

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4.6.4 Operability.- The following tests shall be performed with standard torque measuring equipment equipped with appropriate adapters to accommodate the assembly under test in conjunction with the special testing equipment specified in 4.5.2.3.3.

4.6.4.1 Locking torque (yoke assembly to adapter mount).- With the adapter affixed to the locking knob of the adapter mount, the locking knob shall be rotated through the range specified in 3.5.3. The running torque required to secure each yoke assembly gage to the adapter mount shall not be less than the value specified in 3.5.3.

4.6.4.2 Unlocking torque (yoke assembly from adapter mount).- Perform this test in the same sequence as specified in 4.64.1. The locking knob shall be rotated through the range specified in 3.5.4. The maximum torque required to unlock the locking knob and release each yoke assembly gage from the adapter mount shall not exceed the value specified in 3.5.4.

4.6.4.3 Azimuth knob rotation.- The maximum torque required to rotate the azimuth knob adjustment shall be measured by a standard type torque wrench and appropriate adapter, and shall not exceed the values specified in 3.5.8 when subjected to the temperatures indicated in 3.4.9.

4.6.4.4 Plate rotation.- The minimum and maximum torque required for rotation of Plate C10541125 shall not be less than the values specified in 3.5.9 and 3.5.10 at the temperatures indicated 3.4.9 when measured by means of a standard type torque wrench and appropriate adapter.

4.6.4.5 Leveling adjustment screws.- When measured by a standard type torque wrench with appropriate adapter, the maximum torque required to rotate each leveling adjustment screw shall not exceed the values specified in 3.5.11 when subjected to the temperatures indicated in 3.4.9.

4.6.4.6 Cable assembly.- This test shall be conducted at the extreme low temperature only, as specified in 3.5.12. Utilizing a 1/2 inch mandrel firmly secured in a holding medium, the approximate center of the cable length shall make initial contact on the top surface of the mandrel and equal pressure shall be manually applied at the cable extremes to effect the 270° bend. After ten bends (minimum) at the extreme (low) temperature there shall be no damage evident as a result of a visual examination and a continuity check shall be conducted by means of standard measuring equipment to insure compliance with 3.5.12.

4.6.4.7 Spindle rotation.- The torque on spindle D10541117 shall be determined by utilizing an appropriate adapter with a standard torque wrench having a range of at least 20 inch pounds with an accuracy of ± 1/4 inch pound. Torque shall be determined throughout the excursion specified in 3.5.5 to determine conformance with 3.5.13.

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4.6.5 Orientation.- Perform the following tests with the adapter mount positioned on the special testing equipment as shown in Figure 1. The test equipment shall conform to the requirements of 3.5 and 4.5.2.3.1. The azimuth test fixture and spindle of the adapter mount shall be cross leveled utilizing a 10 second level.

4.6.5.1 Azimuth excursion.- With the azimuth test fixture positioned at zero, rotate the adapter mount spindle to its zero limit stop and slip the azimuth scale of the mount to indicate zero mils in azimuth. Observe through the viewing telescope. Align the intersecting point of the telescope vertical and horizontal reticle lines coincident to the intersecting point of the vertical and horizontal target image lines. View through the microscope and adjust the microscope eyepiece for best focus of the adapter mount azimuth scale. Establish coincidence of a referenced reticle line of the microscope with respect to the adapter mount index. By means of the azimuth knob, rotate the adapter mount through the azimuth excursion specified in 3.5.5 while viewing through the microscope, then sight through the viewing telescope and rotate the azimuth test fixture to re-establish coincidence of the viewing telescope vertical reticle line with respect to the vertical target image line. The azimuth excursion of the adapter mount shall meet the requirements specified in 3.5.5, as measured on the azimuth test fixture scale.

4.6.5.2 Azimuth error.- Perform this test utilizing the zero alignment procedure specified in 4.6.5.1, prior to performing the azimuth error test. The azimuth readings shall be made in 10 mil increments as specified in 3.5.6. While observing through the microscope to the azimuth scale of the adapter mount, turn the azimuth knob of the mount and rotate the spindle to each mil value specified above, as read on the azimuth scale. As each 10 mil position is established on the azimuth scale sight through the viewing telescope and rotate the azimuth test fixture to establish coincidence of the telescope vertical reticle line with respect to the vertical target image line. Record the azimuth test fixture readings at each 10 mil position until the limit stop of the adapter mount is reached. Azimuth error shall not exceed the tolerance specified in 3.5.6, as measured on the azimuth test fixture. Upon reaching the limit stop, record the last 10 mil azimuth scale measurement of the adapter mount for this position. Return the spindle of the adapter mount to the azimuth zero index stop and position the azimuth test fixture to indicate zero mils in azimuth. Establish coincidence of the viewing telescope and target image. While viewing through the microscope rotate the azimuth scale of the adapter mount from the limit stop until the last 10 mil recorded position is superimposed with respect to the reference reticle line of the microscope. Continue testing the remainder of the azimuth scale as noted above. The recorded total azimuth readings through the full excursion of the adapter mount shall meet the requirements specified in 3.5.6, as measured on the azimuth test fixture.

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4.6.5.3 Circular level.- The level bubble shall conform to the requirements specified in 3.5.7 as the yoke assembly is rotated through the azimuth excursion specified in 3.5.7.

4.6.6 Electrical.

4.6.6.1 Continuity.- Standard measuring equipment shall be utilized to determine electrical continuity conforming to the schematic Drawing F10541111.

4.6.6.2 Insulation resistance.- Insulation resistance shall be measured between mutually insulated parts by means of a megohm bridge or other approved method.

4.6.7 Connector interface requirement.- The plug assembly D10541118 of the adapter mount shall be engaged to a test connector with pertinent interfaces conforming to requirements of Drawing B11730226. Continuity between the plug and test connector shall be indicated when fully engaged.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10541111. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The adapter mount is used to support and level the Range Finder, Laser AN/GVS-3 and contains the azimuth scale and azimuth control mechanism.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.
(e) Order of document precedence.
(f) Provisions for First Article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A254

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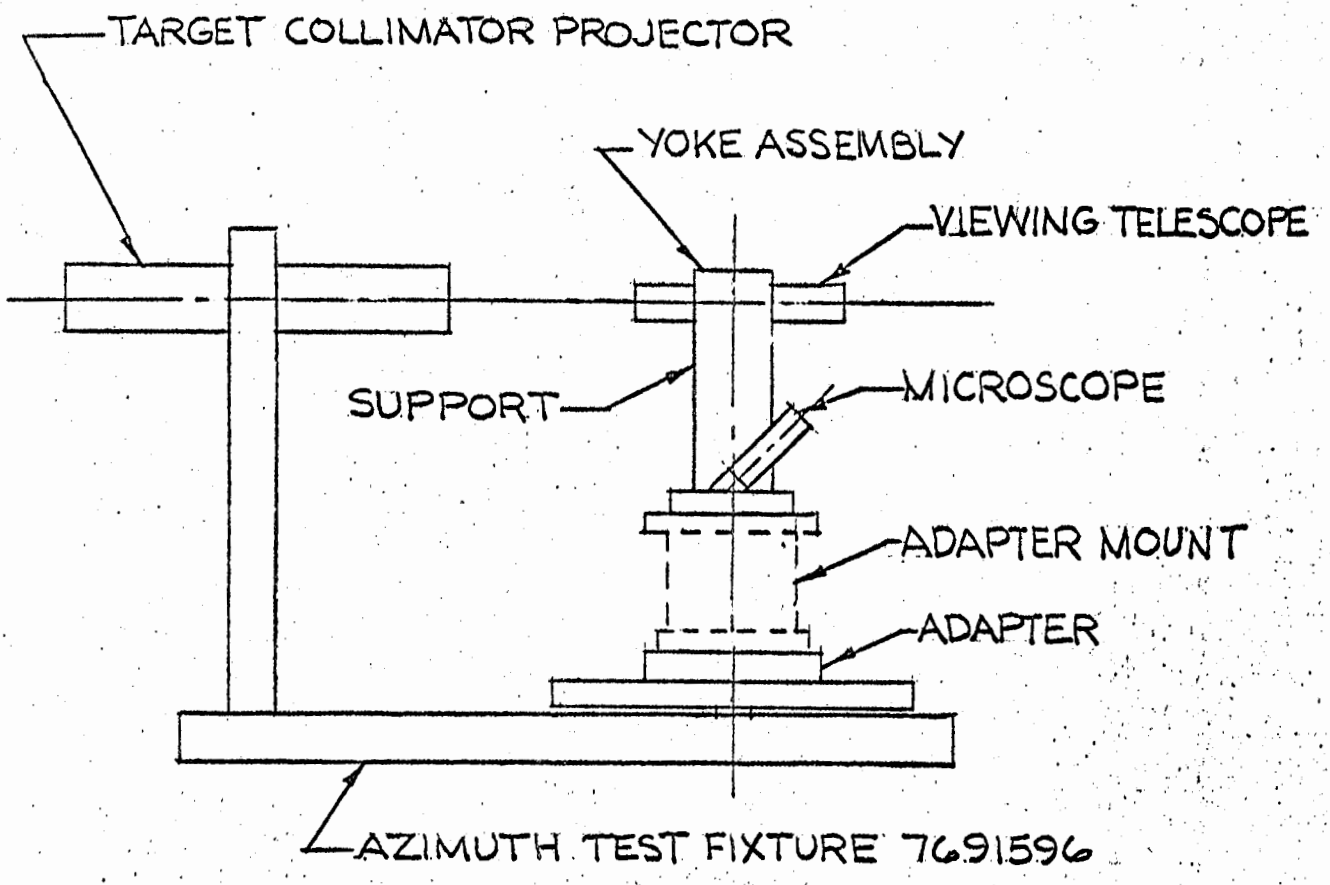


FIGURE 1

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